

SCITRAM: Single-cell Integrated transcription regulation modeler

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Tutorial

1. Introduction

Accessing to the molecular cell heterogeneity thanks to the development of single-cell omics technologies is revolutionizing our understanding of complex living systems. While generating such type of data is becoming a streamlined process, data processing remains to be possible either by command-line driven tools, or commercial solutions. Furthermore, single-cell transcriptomics data processing is usually reduced to cells stratification on the grounds of gene expression signatures, while aspects like deconvolving the various gene regulatory programs responsible for cells heterogeneity are not classically addressed with the available tools, or performed under different data processing platforms.

Herein we introduce SCITRAM, a computational solution able to (i) process multiple single-cell transcriptomic files at once, (ii) perform dimensionality reduction while retaining the origin of cells retrieved on the obtained clusters, (iii) compute differential gene expression profiling among stratified clusters; (iv) annotate clusters based on the use of user-provided cell-type gene association information; (v) reconstitute gene regulatory networks per cluster by using transcription factors-target gene relationships; and (vi) classify transcription factors as master drivers defining cluster differences retrieved within single-cell transcriptomics readouts.

All the aforementioned steps are possible within an intuitive user-friendly environment, making of SCITRAM a unique platform providing a large number of data processing options. Furthermore, SCITRAM distribution is meant to avoid tedious tools/dependencies installations, relying in a direct executable distribution format for both windows and Mac, in addition to the regular Linux distribution.

To illustrate the performance of SCITRAM we illustrate in this tutorial outputs issued from the analysis of single-cell transcriptomics data concerning CD8+ T cells collected in the context of solid tumors treated with CAR-T cells. This data are part of the main publication describing SCITRAM and were generated by the team of [Sonia Guedan](#) (Institut d'Investigacions Biomèdiques Agustí Pi i Sunyer, Spain) within the frame of the European collaborative project [T-FITNESS](#).

2. CITRAM installation

Windows / OSX Installation

We provide 3 versions, one for Microsoft windows (working under versions 10 and 11), one for OSX (Mac) and one python script for Linux and other systems.

To use SCITRAM under **Windows** system you have to:

- 1) Download the SCITRAM_vxx.zip folder following the next link:
<https://github.com/SysFate/SCITRAM>
- 2) De-compress the file.
- 3) Double-click on the SCITRAM_vxx.exe executable file.

Nom	Modifié le
_internal	30/08/2026 16:56
ExtraFiles	30/08/2026 16:56
GRN_collection	30/08/2026 16:56
image	30/08/2026 16:56
Markers_list	30/08/2026 16:56
Network	30/08/2026 16:56
SCITRAM_3.64.exe	30/08/2026 16:56

Caution: Keep the directories in the same place that the executable file (.exe), since several files required for the tool are available in the same place.

SCITRAM under OSX:

In preparation

Linux installation

1. Go to <https://www.anaconda.com/download>
2. Register your account or simply skip it.
3. Chose the Linux distribution and select the 64-bit (x86) installer



Alternatively, you can go directly to download the file in your shell window, type:

```
wget https://repo.anaconda.com/archive/Anaconda3-2024.10-1-Linux-x86_64.sh
```

4. Open your shell window, browse to “Downloads” and type :

```
sh Anaconda3-2024.10-1-Linux-x86_64.sh
```

Follow the instructions.

Caution: Conda versions can change over time; but what is essential is that the file name of the required version should resemble to “Anaconda3-20XX.XX-X-Linux-x86_64.sh”

5. Go to our github: <https://github.com/SysFate/SCITRAM> and download the files SCITRAM_vxx.py and the file SCITRAM_requirements.yml

6. Create and activate new environment typing

```
conda create -n SCITRAM -f SCITRAM_requirements.yml
conda activate SCITRAM
```

7. Go to the SCITRAM directory and run the python script:

```
python SCITRAM_vxx.py
```

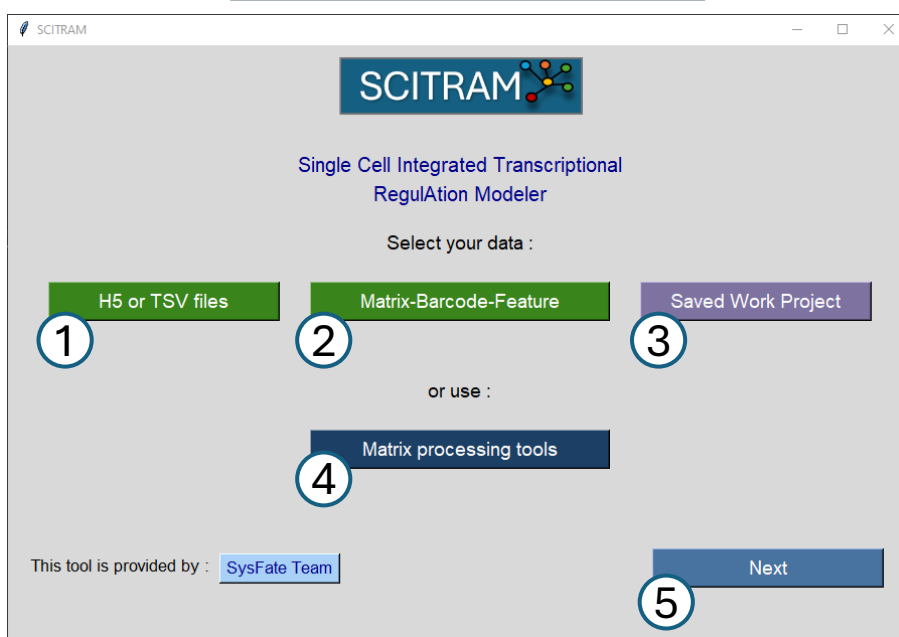
3. Analyzing single-cell transcriptomics data with CITRAM

3.1. Reading Single Cell Files

SCITRAM was developed to read the most common files: hierarchical data format (.H5), matrix in plain text delimited by tabs (.tsv), and files in format [matrix.mtx - barcodes.tsv, - features.tsv]. In addition, users can upload multiple files if wished.

Datasets need to be preprocessed for converting the raw sequencing reads and base quality scores generated from individual cells (FASTQ format) into file formats hosting the gene identity retrieved per cell, as well as the number of counts per cell. For example, 10X Genomics provide the tool Cell ranger, which will generate H5 or matrix-barcodes-features files using the genome pre-build for this platform. Some sequencing platforms already provide this files, but if you have only the “fastq” files you need to do this pre-processing for example in [Cloud Analysis | 10x Genomics](#). or [Installing Cell Ranger - 10x Genomics](#). Repositories like NCBI GEO provide in most cases files in SCITRAM compatible formats.

Upload files window (1)



Select your H5 or plain text in **(1)**, users can select many files as they want (that depends on the capacity of your computer). For table with counts for the plain text file, the gene names need to be placed in the first column, and the “name of cells” (barcodes) need to be in the first row.

Gene names should be the “Gene symbol” approved by the HGNC, this is for example actin beta: “ACTB” and Not: “ENSG00000075624”. For the mouse genome the accepted format should be “Actb”. Text separators could be tab (.tsv) or comma (.csv) delimited.

In **(2)** the user can upload the data with the system of 3 files (matrix, barcodes and features).

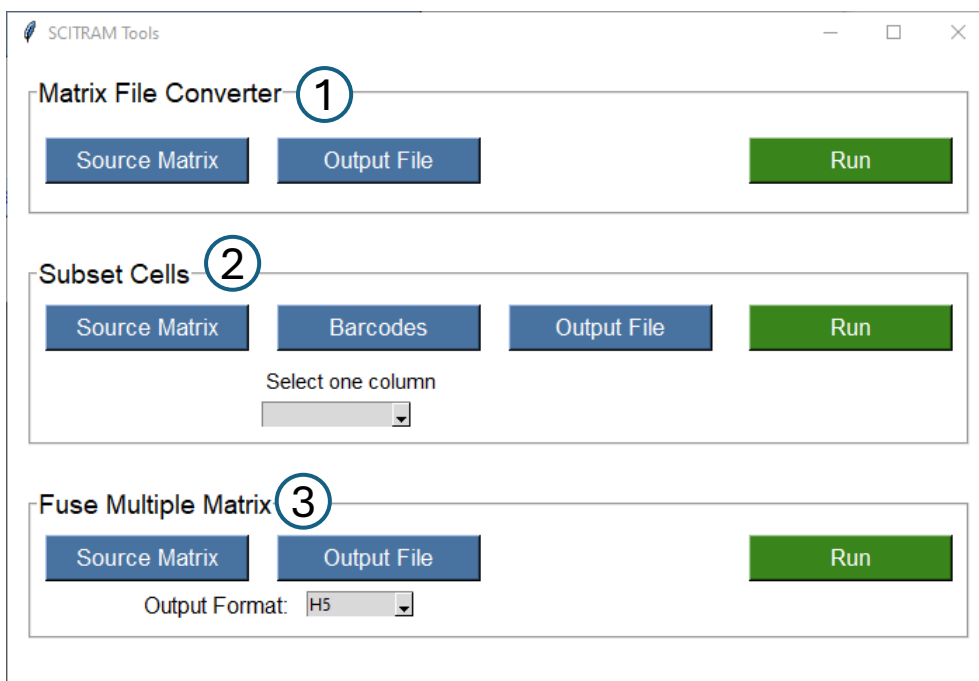
Caution: For uploading multiple datasets at once, users need to select the directory where folders holding the various datasets are retrieved. The name of the folder correspond to the identity of the dataset, while each of the files within the folder should be named only with the corresponding extension (Example: dataset1/ matrix.mtx, barcodes.tsv, features.tsv; dataset2/ matrix.mtx, barcodes.tsv, features.tsv; etc.)

In **(3)** users have the possibility to upload a previously saved work.

In **(4)** users will find some tools for pre-processing the data

To continue setup the analysis, click in “Next” **(5)**.

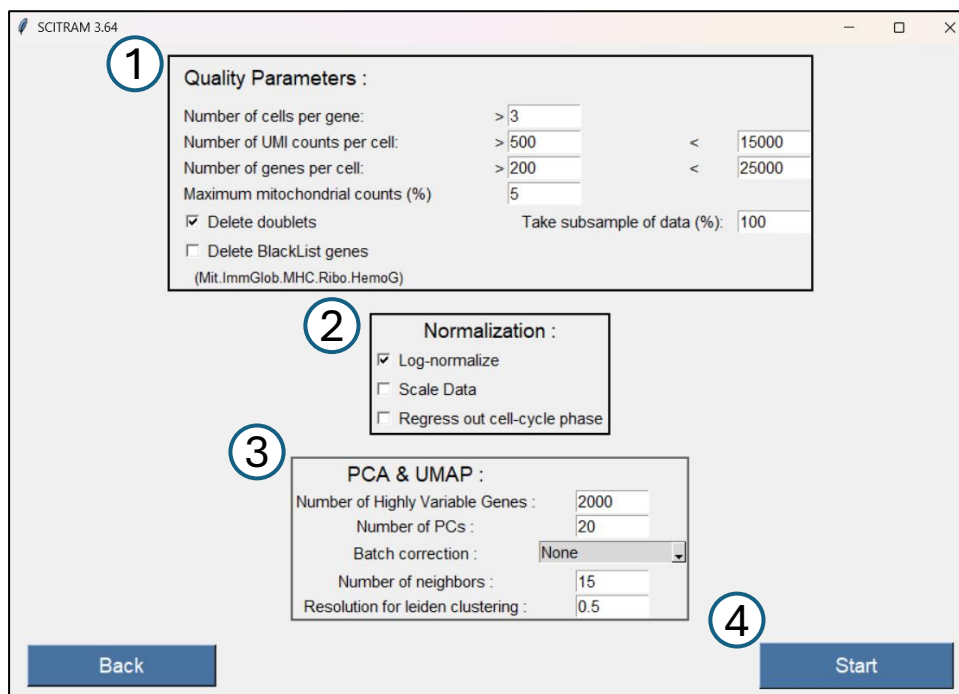
3.1.1. Optional: Data pre-processing tools



The image shows a screenshot of the 'SCITRAM Tools' window. It contains three distinct tool panels, each with a numbered circle (1, 2, and 3) indicating their sequence.
Tool 1: Matrix File Converter - Features two input buttons labeled 'Source Matrix' and 'Output File', and a green 'Run' button.
Tool 2: Subset Cells - Features three input buttons labeled 'Source Matrix', 'Barcodes', and 'Output File', a green 'Run' button, and a dropdown menu labeled 'Select one column'.
Tool 3: Fuse Multiple Matrix - Features two input buttons labeled 'Source Matrix' and 'Output File', a green 'Run' button, and a dropdown menu labeled 'Output Format' with 'H5' selected.

SCITRAM counts with a group of tools to help in data pre-processing before to start the main analysis. For these 3 tools you need to select your sample, select the output file name and directory, and simply run. In all cases the input and output could have the format “.h5” or “.tsv”. In **(1)** you can interconvert files between 2 types of formats described in the section above. “.H5” $\Leftrightarrow$ “.tsv, .tsv, .txt” . You can subset cells in **(2)** using your matrix and a tab delimited file with the barcodes (cell names), for this you need to select the column that contain the barcodes. You can save time in the main process if you fuse your multiple files in **(3)**.

3.2. Define parameters for data processing



SCITRAM 3.64

1 Quality Parameters :

Number of cells per gene: > 3

Number of UMI counts per cell: > 500 < 15000

Number of genes per cell: > 200 < 25000

Maximum mitochondrial counts (%) 5

☒ Delete doublets Take subsample of data (%): 100

☐ Delete BlackList genes
(Mit.ImmGlob.MHC.Ribo.HemoG)

2 Normalization :

☒ Log-normalize

☐ Scale Data

☐ Regress out cell-cycle phase

3 PCA & UMAP :

Number of Highly Variable Genes : 2000

Number of PCs : 20

Batch correction : None

Number of neighbors : 15

Resolution for leiden clustering : 0.5

4

Back Start

With the aim of addressing expectations of naïve but also advanced users, SCITRAM proposes a by-default parameter setup, but it gives the flexibility to the user to modify them if wished.

In panel **(1)** you can use different parameters to filter your samples in terms of cells, genes or counts:

- **Number of cells per gene:** avoid to have genes present in a few number of cells.
- **Number of reads per cell:** filter out cells with few counts, but also highest amounts of counts notably because could be consequence of doublets, triplets in the same encapsulated droplet.
- **Filter cells (% mito counts):** Previously was calculated that one cell have around 5 % of counts from mitochondrial genes. This filter is of major relevance in cases in which, for instance, cells are known to be in mitosis, where there is more active mitochondria. Take in account that some experiments like Multiomes require to separate the nucleus. Thus, in the case of multiome RNA + ATAC-seq , it is advised to use the filter 10%.
- **Delete doublets:** This process take in account the expected number of counts respect to the number of cells and the saturation of the system (data provided by 10x). This parameter requires more processing time because basically it runs the process twice.
- **Delete Blacklist genes:** You can get out of the analysis genes that are irrelevant for your study and/or have very high levels and/or variables in the cells. For example, this is the case of immunoglobulins, hemoglobins, ribosomal genes. The list of genes that are concerned by this cleaning step is available in the file “Human_blacklist_genes_Ensembl115.tsv” or “Mouse_blacklist_genes_Ensembl115.tsv” inside the “ExtraFiles” directory. Users have the possibility to modify this file if wished.

In panel (2) you can setup the normalization parameters through the cells:

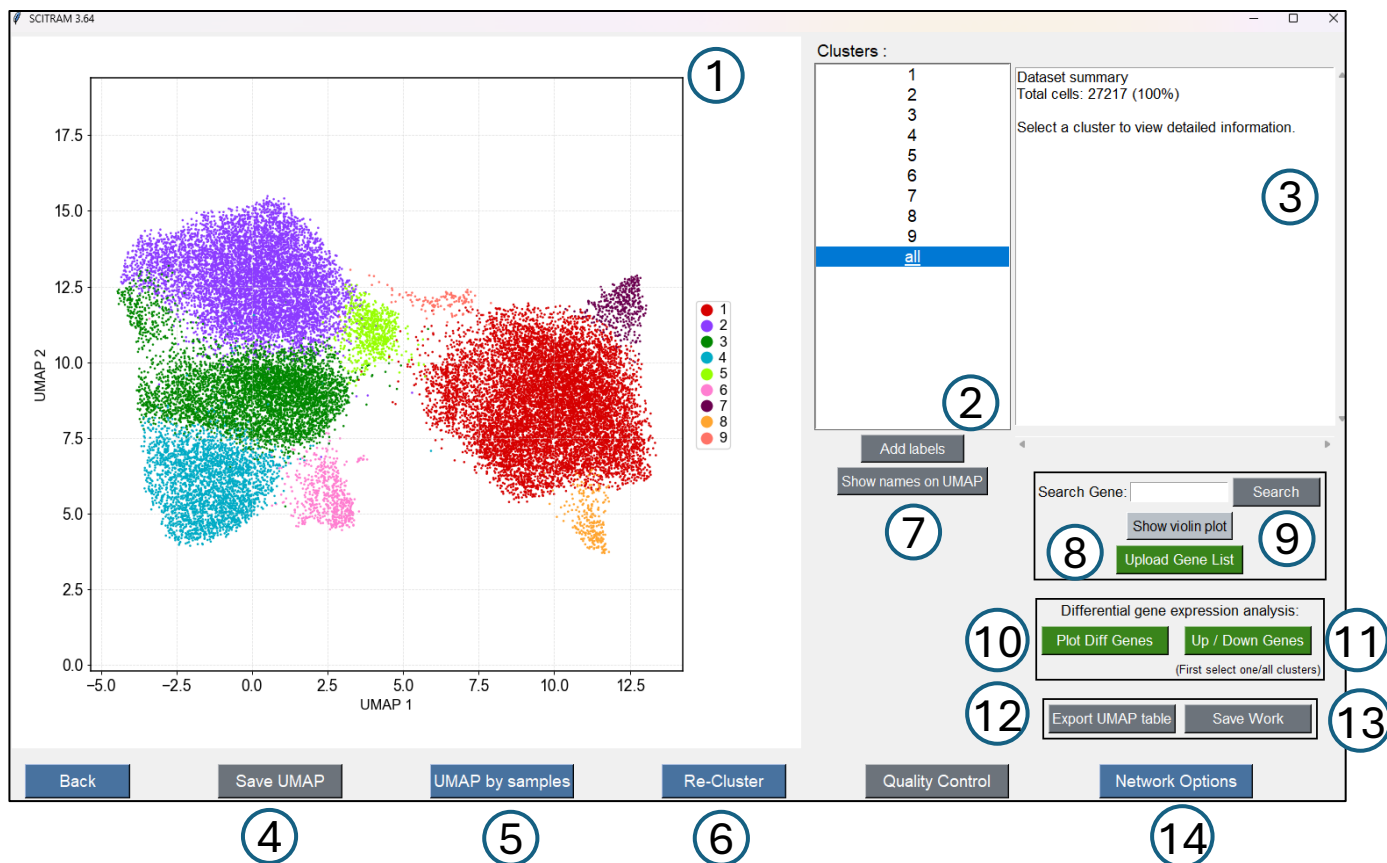
- **Log-Normalize:** this is a normalization used by default in different packages like Scanpy and Seurat. It uses the modified logarithmic normalization to avoid outliers (intra-samples).
- **Scale data:** this method will provide a relative number of counts for each gene through the cells, that's is zero for the media of the gene and the difference of the values in log 2 scale.
- **Regress out cell-cycle phase:** The normalization takes in account the expression of some genes like the cyclins. This can help in experiments where you assume that your cells have the same levels of cell division. If is not the case and you are searching this differences, is better to avoid this process.

Finally, for the PCA and UMAP setup (3), you can modify the way of how information you should use to separate the cells, how intensive should be the analysis and how to group the cells.

- **Number of highly variable genes:** When you have many cells, the capacity to analyze all the genes to classify these cells could take several hours or crash the computer, even could generate errors, we assume that the most variables genes are around 2,000, maybe if you need to be mor exhaustive you can increase.
- **Number of PCAs:** you can increase if you want to be more exhaustive in your analysis, but certainly when you increase the number of PCs there are no big differences in the number of clusters obtained.
- **Number of neighbors:** To group your cells you can assume that your cells are very connected between them (high values) or less connected (lower values).
- **Resolution of Leiden clustering:** maybe the most important of these parameters, because you can increase or decrease the number of clusters. However, because this is a non supervised analysis, you can't generate a specified number. To obtain more clusters you can increase this parameter, and notably once you have already performed a first analysis with the “by-default” parameter.
- **Batch correction:** In case users would have multiple datasets, SCITRAM counts with the following batch correction modules: (i) Harmony (Korsunsky et al; Nat. Methods 2019); (ii) BBKNN (Polanski et al; Bioinformatics 2020); (iii) Scanorama (Hie et al; Nat. Protocols 2024).

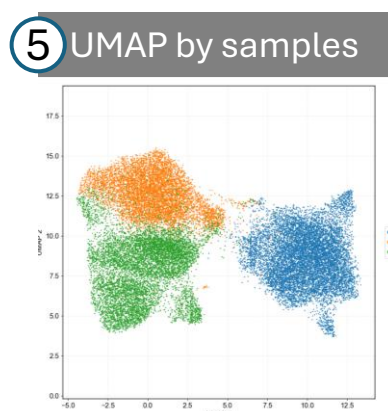
3.3. Interacting with the processed data via the main window display

This main display provide many ways to visualize your data, both by graphic displays as well as in table format.

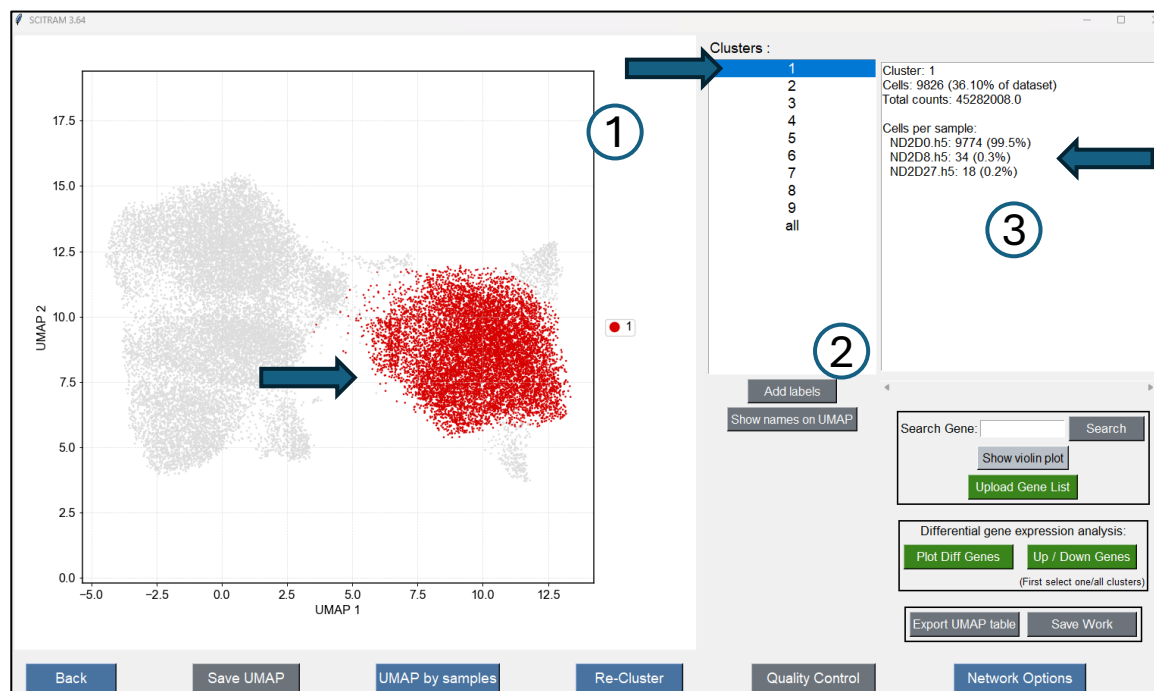


In (1) you can visualize the UMAP in different ways, modified by other analysis. Any time during the analysis, users can save the displayed graphics with the button “Save UMAP” (4).

SCITRAM allows to visualize cells stratification based on the datasets of origin (5):



By selecting a given cluster **(2)**, the user can get the number of cells per cluster as well as its stratification based on the different uploaded data **(3)**. Users can also add a label to the clusters **(2)**.



6

In cases in which a further stratification is required for a given cluster, SCITRAM provides a re-clustering option **(6)**

Re-clustering window

6 Re-Cluster

Local Re-Clustering

Select one cluster:

1

PCA & UMAP:

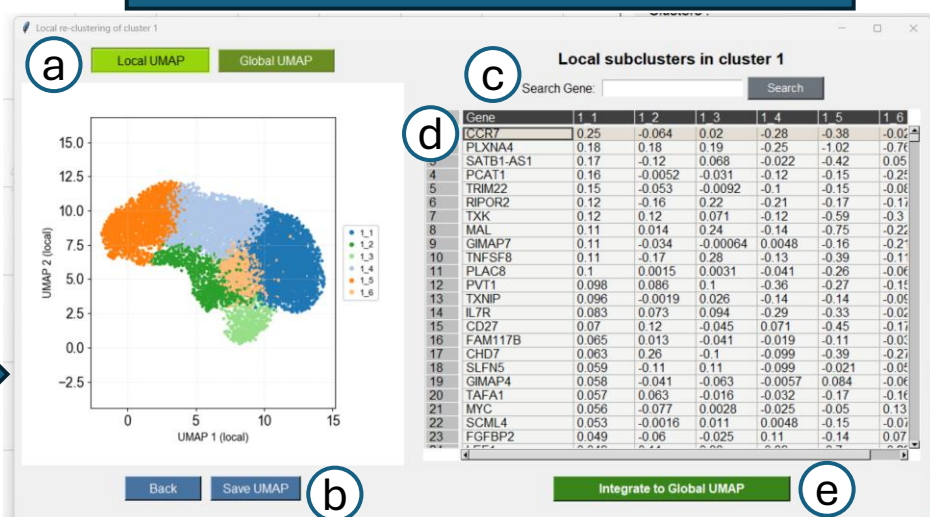
Number of Highly Variable Genes: 1500

Number of PCs: 15

Number of neighbors: 10

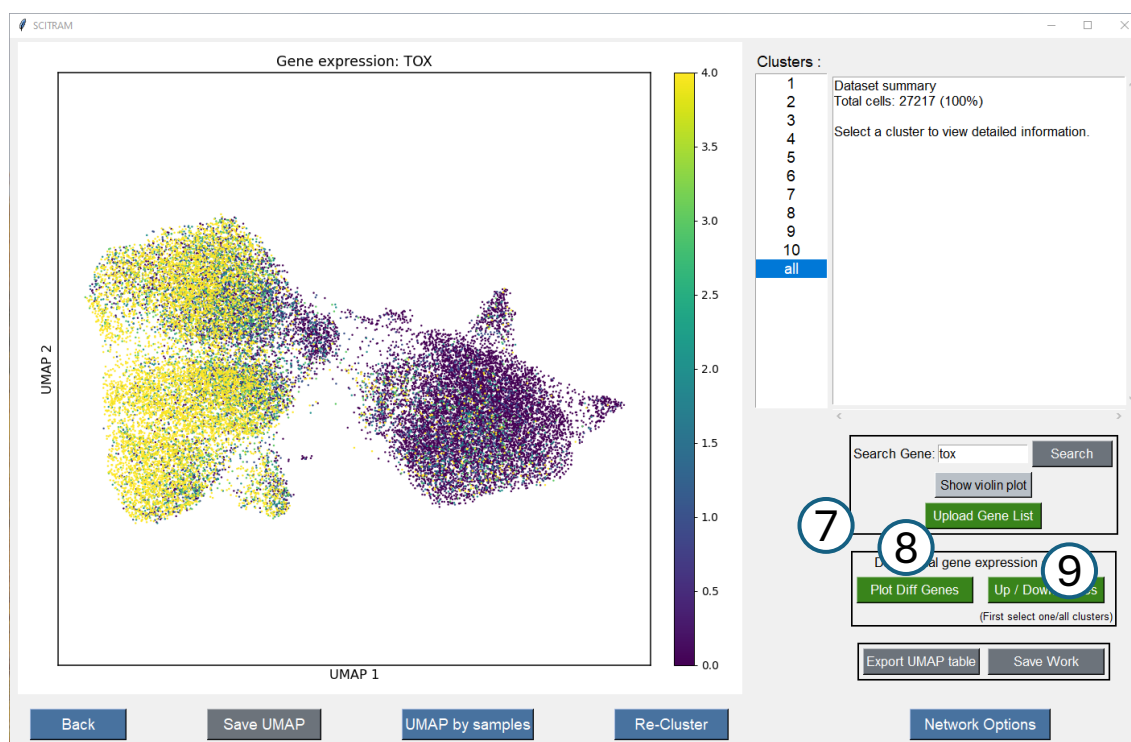
Resolution for leiden clustering: 0.4

Re-Clustering



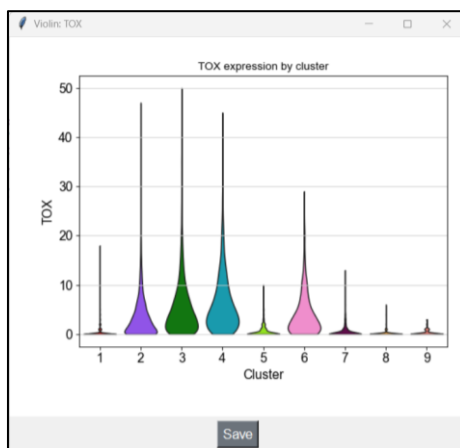
You can see how the cells are clustered in a local or global way (upper buttons) **(a)** and you can see which genes are expressed per subcluster in **(c)**. The table in panel **(d)** is interactive, which allows to copy paste its content for instance in an excel file. The re-clustered map can be saved as image **(b)**, and it can also be integrated within the global UMAP for proceeding with the analysis **(e)**, or it could be saved as a separate object.

SCITRAM allows to visualize the expression levels of a given gene by using a query panel (7).



Gene expression levels can also be displayed in a violin plot format (8):

Violin plot
supplementary
window



Example of a gene
markers annotations table

Cell_Markers_list.tsv		
Fichier	Modifier	Affichage
marker_type	gene	
Effector_markers	CST7	
Effector_markers	GZMK	
Effector_markers	GZMA	
Effector_markers	NKG7	
Effector_markers	IFNG	
Effector_markers	PRF1	
Effector_markers	GZMB	
Effector_markers	GNLY	
Exhaustion_markers	PDCD1	
Exhaustion_markers	TOX	
Exhaustion_markers	CXCL13	
Exhaustion_markers	TIGIT	
Exhaustion_markers	CTLA4	
Exhaustion_markers	TNFRSF9	

3.4. Annotating clusters by using gene marker tables

Furthermore, when larger list of genes are wished to be analyzed users can upload a Gene list in (9) where the first column corresponds to a “cell marker type” annotation (or any other annotation of interest) and the second to the genes ID of interest (official gene symbols) see example below.

Caution: This table needs to count with the headers “marker_type” and “gene”.

Once uploaded, the list of genes and its corresponding annotation is displayed in a table where the frequency of expression of each gene per cluster is displayed and could be saved as excel format (c):

9 Query list supplementary window

7 Search Gene: Search

8 Show violin plot

9 Upload Gene List

10 Differential gene expression analysis:
Plot Diff Genes Up / Down Genes (First select one/all clusters)

12 port UMAP table Save Work

14 Network Options

SCITRAM

All Clusters

	marker_type	gene	cluster 1	cluster 2	cluster 3	cluster 4	cluster 5	cluster 6	cluster 7	cluster 8	cluster 9
6	Effector_markers	PRF1	0.87	2.57	1.52	2.01	1.37	1.12	1.45	0.47	0.59
7	Effector_markers	GZMB	23.27	29.32	10.20	12.53	18.06	5.85	59.67	13.11	8.73
8	Effector_markers	GNLY	2.71	24.14	40.20	43.79	36.75	38.64	2.43	1.41	7.75
9	Exhaustion_markers	PDCD1	0.0069	0.33	0.19	0.084	0.091	0.16	0.064	0.0074	0.038
10	Exhaustion_markers	TOX	0.36	3.65	5.36	6.32	0.57	3.96	0.46	0.14	0.23
11	Exhaustion_markers	CXCL13	0.0012	0.42	1.99	0.52	0.068	0.79	0	0.0037	0.06
12	Exhaustion_markers	TIGIT	0.12	1.98	2.53	2.29	0.96	1.72	0.1	0.22	0.31
13	Exhaustion_markers	CTLA4	0.026	0.94	1.29	1.89	0.38	0.83	0.038	0.015	0.11
14	Exhaustion_markers	TNFRSF9	0.018	2.81	1.08	1.18	0.19	0.98	0.63	0.0074	0.34
15	Exhaustion_markers	HAVCR2	0.57	1.66	1.91	0.67	1.36	1.41	1.45	0.53	0.2
16	Exhaustion_markers	LAG3	0.27	2.16	1.25	1.10	1.09	0.8	0.64	0.35	0.44
17	Exhaustion_markers	NR4A1	0.016	0.51	0.34	0.27	0.29	0.59	0.16	0.0074	0.09
18	Tex_GZMKpos	CD28	0.1	0.16	0.032	0.28	0.05	0.08	0.12	0.089	0.015
19	Tex_GZMKpos	EOMES	0.0084	0.01	0.0064	0.0011	0.016	0.0051	0.026	0.0074	0.007
20	Tex_GZMKpos	CCR5	0.099	0.17	0.12	0.11	0.24	0.09	0.047	0.052	0.038
21	Tex_GZMKpos	CCL4	0.81	9.69	6.45	2.87	11.36	4.26	27.74	0.54	2.81
22	Tex_GZMKpos	CCL5	6.12	5.98	12.31	5.96	9.48	11.16	5.80	10.11	2.53
23	Tex_GZMKpos	CD200R1	0.058	0.018	0.14	0.026	0.11	0.16	0.031	0.019	0.015

(No Title)

Show on UMAP

☒ Include Heatmap

Export table

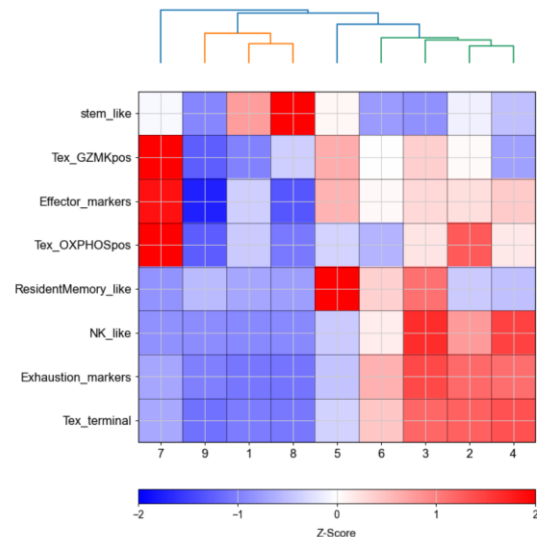
Heatmap all markers

All markers heatmap supp window

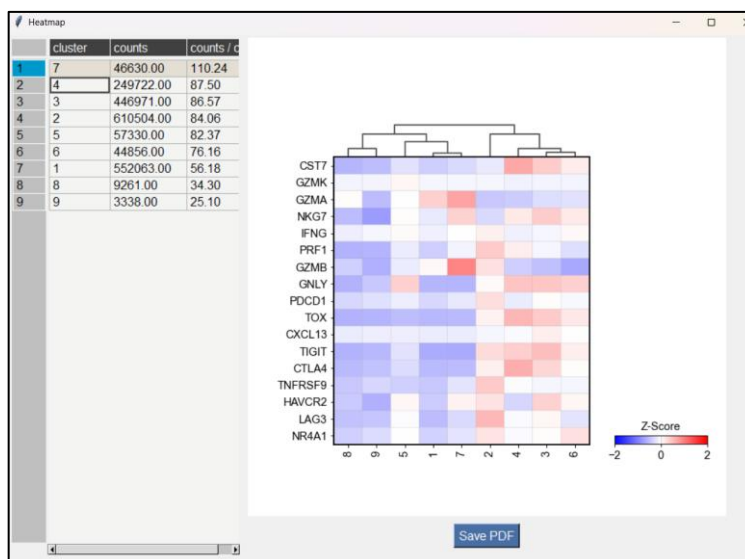
By clicking on **(d)** SCITRAM provides a global view of the enrichment of the queried cell markers across the different clusters.

In this case, you can see that exhaustion markers are preferentially enriched on clusters 6,3,2 and 4; effector markers in cluster 7 and stem-like markers in cluster: 8.

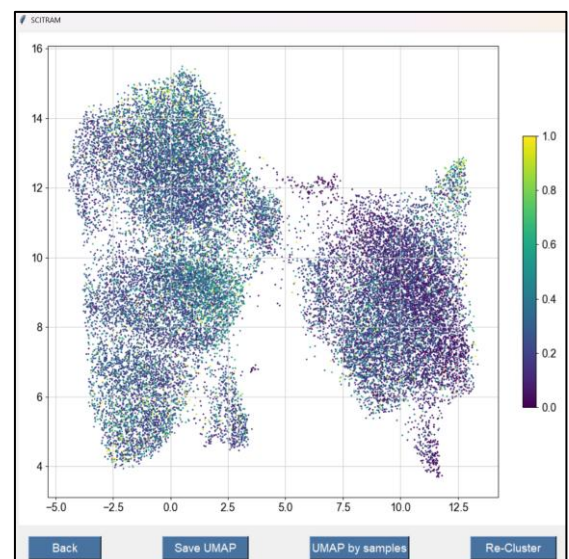
We can also select some rows on the table to obtain a view of the selected genes per cluster heatmap and their display within the main UMAP **(a)** and **(b)**



Selected genes heatmap



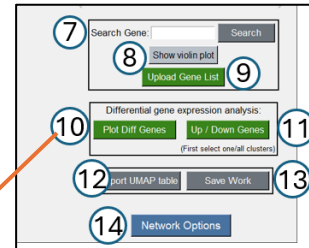
Rel. gene markers enrichment



3.5. Perform a differential expression analysis between cell clusters

To perform a differential expression analysis between clusters, SCITRAM allows to select the clusters of interest. In this example, cluster 3 (97% cells issued from day 27 in CAR-T cells treatment;) vs cluster 2 (96% cells from day 8) are compared.

Main window: selected clusters 3 and 2



Plot Diff Genes

Differential Expression Analysis

Reference cluster(s):
1
2
3
4
5
6
7
8
9
10

Target cluster(s):
1
2
3
4
5
6
7
8
9
10

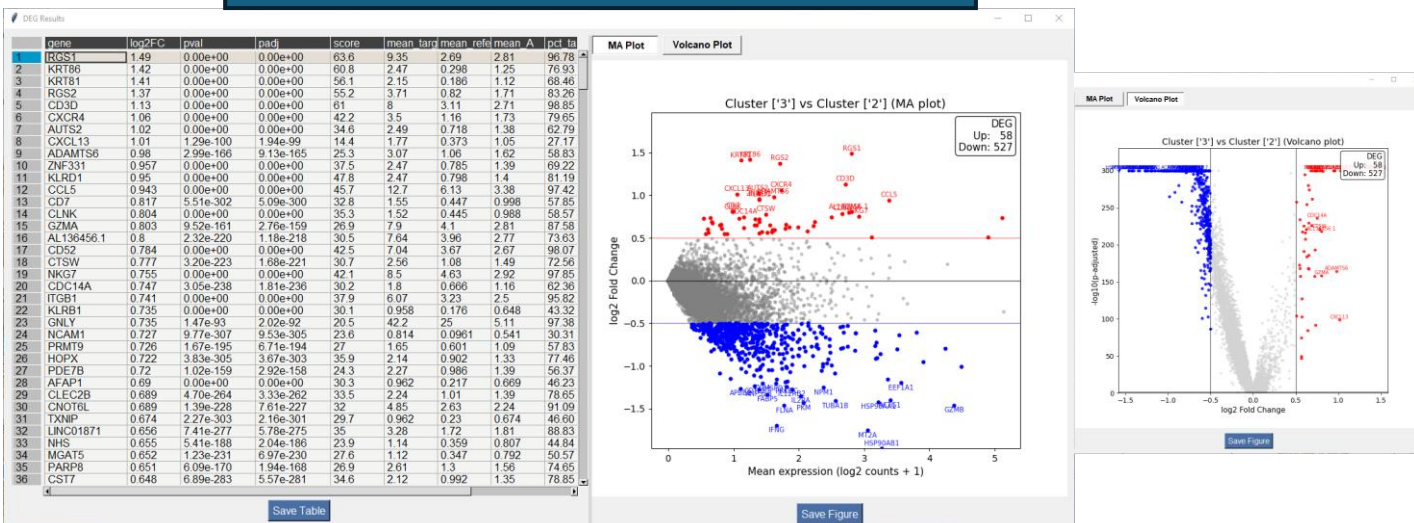
LFC > 0.5
p-value < 0.05
Top N genes: 20

Analyze

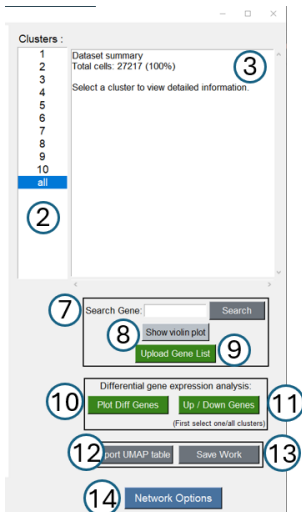
When selecting “Plot Diff Genes”, a new window will be open where users can select the clusters to compare, the log2 fold-change (LFC) and the P-value confidence threshold, as well as the number of top-ranked genes to display in the analysis.

The analysis will provide a table that can be saved in tsv or excel formats, and a panel where either a “Minus versus Average” (MA) plot, or a “volcano” plot can be displayed. In addition to the displayed name of genes based on the previous parameter, users can read the gene name of other points by passing the mouse pointer.

Plot Diff Genes display



SCITRAM can also provide a direct view of all differential gene expression levels by clicking on “Up/Down Genes” **(11)** across all clusters at once if « all » clusters are selected; or for a given selected cluster in **(2)**:



Clusters :

Dataset summary
Total cells: 27217 (100%)

Select a cluster to view detailed information.

1
2
3
4
5
6
7
8
9
10
all

7 Search Gene: Search

8 Show violin plot

9 Upload Gene List

Differential gene expression analysis:

10 Plot Diff Genes

11 Up / Down Genes
(First select one/all clusters)

12 Sort UMAP table

13 Save Work

14 Network Options

11 Up / Down Genes

SCITRAM

Mean LFC across all clusters

gene	cluster 1	cluster 2	cluster 3	cluster 4	cluster 5	cluster 6	cluster 7	cluster 8	cluster 9	mean_LF
1 SOS1	1.30	-0.79	-1.28	-1.31	-0.62	-1.39	0.48	1.78	-2.02	-0.43
2 IFITM2	1.12	-1.10	-0.85	-1.63	0.014	-1.08	1.19	0.83	-1.07	-0.29
3 PPIB	0.99	-0.069	-0.67	-0.82	-0.63	-0.97	1.18	0.47	-1.17	-0.19
4 GZMA	0.99	-1.47	-0.64	-1.29	0.045	-0.57	1.65	0.29	-2.03	-0.34
5 CD52	0.95	-0.89	-0.034	-0.22	-0.45	-0.61	0.91	0.3	-1.12	-0.13
6 IKZF1	0.93	-0.33	-0.77	-0.51	-0.44	-0.39	0.77	0.73	-1.28	-0.14
7 CALR	0.93	0.12	-0.59	-0.71	-0.54	-0.82	1.05	0.33	-1.13	-0.15
8 EBP	0.9	-0.098	-0.52	-0.55	-0.23	-0.64	0.68	0.29	-0.59	-0.084
9 S100A6	0.89	0.014	-0.91	-0.85	0.039	-0.73	0.57	0.52	-0.54	-0.11
10 MLLT3	0.89	-0.33	-0.57	-0.41	-0.66	-0.51	0.53	1.10	-1.57	-0.17
11 CTSC	0.88	-0.15	-0.65	-0.056	-0.21	-0.4	0.66	0.1	-0.89	-0.079
12 BCL2	0.88	-0.63	-0.1	-0.0057	-0.54	-0.083	0.48	0.56	-1.83	-0.14
13 THEMIS	0.87	-0.62	-0.42	-0.12	0.00022	-0.041	0.022	0.83	-1.73	-0.13
14 HSPA5	0.87	-0.17	-0.75	-0.83	-0.74	-0.76	1.27	0.45	-0.9	-0.17
15 HSPA8	0.85	-0.22	-1.18	-1.22	-0.96	-0.024	1.17	0.71	-1.04	-0.21
16 DPP4	0.83	-0.28	-0.89	-0.97	-0.15	-0.7	0.82	0.84	-0.81	-0.15
17 HTATSF1	0.82	-0.33	-0.53	-0.51	-0.41	-0.6	0.91	0.42	-0.68	-0.1
18 PLXNA4	0.82	-0.49	-0.4	-0.32	0.031	-0.37	0.52	0.18	-0.46	-0.056

Export table

11 Up / Down Genes

SCITRAM

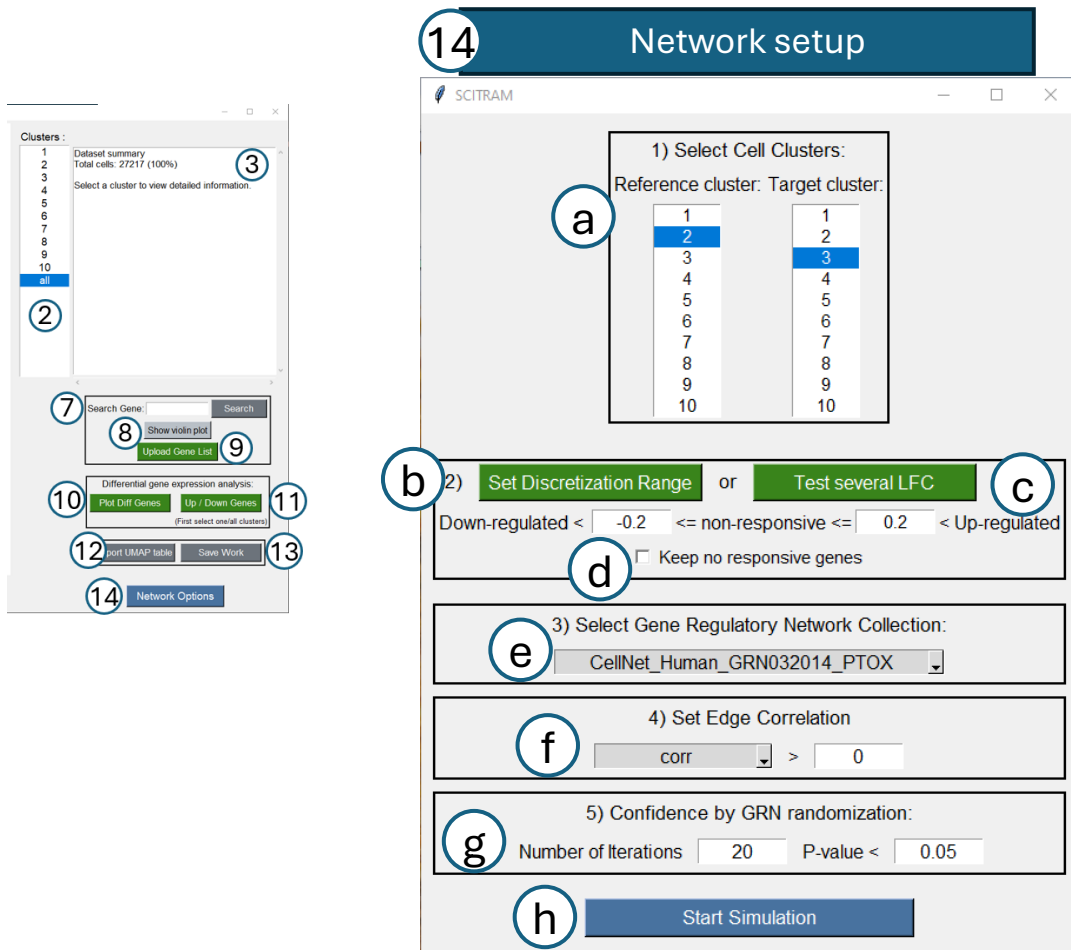
Cluster 2

gene	lfc
1 AHI1	1.81
2 RBPJ	1.57
3 LYST	1.47
4 APBB2	1.42
5 PAM	1.41
6 MT2A	1.34
7 DUSP4	1.32
8 FLNA	1.31
9 TNFRSF9	1.29
10 ATP8B4	1.28
11 AL138720.1	1.28
12 PDE3B	1.26
13 MTSS1	1.25
14 TIAM2	1.24
15 FABP5	1.22
16 IFNG	1.21
17 CDK6	1.20
18 PFN1	1.20

Export table

3.6. Reconstructing gene regulatory networks from cell clusters

SCITRAM has been designed for reconstructing gene regulatory networks (GRNs) from differential gene expression readouts assessed between selected clusters. Hence, when clicking Network Options (14) in the main window, you can set up the network analysis. The following parameters are made available by SCITRAM for enhancing GRN reconstruction:



The screenshot shows the SCITRAM main window with the 'Network Options' tab selected. The interface includes a 'Clusters' list on the left, a 'Dataset summary' section, and a 'Differential gene expression analysis' section. The 'Network Options' section is highlighted with a blue header and contains several sub-sections for configuring the GRN reconstruction process.

1) Select Cell Clusters: This section allows users to select a reference cluster and a target cluster for differential gene expression analysis. The clusters are listed in two columns, with the target cluster highlighted in blue.

2) Set Discretization Range or Test several LFC: This section provides options for defining the fold-change (LFC) range threshold. The 'Set Discretization Range' option is selected, showing a range from -0.2 to 0.2. The 'Test several LFC' option is also available.

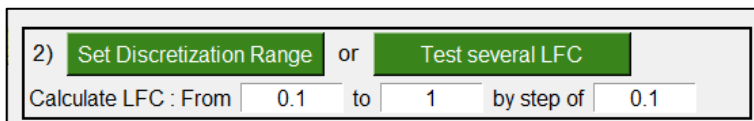
3) Select Gene Regulatory Network Collection: This section allows users to select a gene regulatory network collection from a dropdown menu. The selected collection is 'CellNet_Human_GRN032014_PTOX'.

4) Set Edge Correlation: This section allows users to set the edge correlation threshold. The selected correlation is 'corr' and the threshold is '> 0'.

5) Confidence by GRN randomization: This section allows users to set the number of iterations and the P-value threshold. The selected number of iterations is 20 and the P-value threshold is 0.05.

Start Simulation: A button to initiate the simulation process.

- a)** Selection of the target and the reference cluster for the differential gene expression analysis.
- b)** Define the fold-change (in log2 scale) range threshold to be used for defining differentially expressed genes. The choice of the LFC is important because it is going to define the number of genes that are going to participate at the reconstruction of the gene regulatory maps.
- Caution:** Keep in mind that the log2 fold-change (LFC) provided by default (0.5) is only a suggestion, and requires to be evaluated for obtaining optimal GRNs.
- c)** SCITRAM provides an automated processing for testing several LFCs at once.



This close-up shows the 'Test several LFC' option selected. It includes a 'Calculate LFC' section with a range from 0.1 to 1 by step of 0.1.

- d)** Selecting this option, users can use keep genes with lower LFC than the selected range. This option allows to interrogate for the role of transcription factors that do not present changes in expression levels between compared clusters.

e) Select a regulon collection, composed by TF-TG relationships collected from the public domain. Currently we provide datasets available from CellNet ([Cahan et al; Cell 2014](#)) to analyze Human or Mouse samples. If you want to add other collections, simply add a file in the “GRN collection” directory.

Caution: Regulome files should present the columns “Target Gene”, “Transcription Factor” and a given association “score”.

f) Filter for Edges: This value depends on the GRN collection. In CellNet it corresponds to the correlation parameter “corr” and allows to filter edges with lower correlation values than corresponding threshold (symmetric threshold)

g) This parameter allows to randomize the TF-TG connections to evaluate whether the network connectivity is relevant for the system. You can select the number of randomization events (iterations) and the accepted P-value.

h) Run the process could take several minutes depending of the LFC selected and the number of iterations, a new window will appear finishing the analysis, the main network window.

To help with the choice of the LFC, SCITRAM allows to explore the consequence of various LFC thresholds. When clicking on “Test several LFCs” (**c**) a table containing the LFC threshold in use is provided, the number of differentially expressed genes (DEG), number of Nodes (genes in a network), number of edges (connections between genes). Furthermore, the “Density” metrics (making reference to the level of edge connectivity with the GRN) and number of components, indicative of the number of obtained networks. A component value of 1 means that there is only one network, while values higher than 1 indicates the presence of multiple satellite networks. The row in blue corresponds to a network with higher density and one point before to have satellite networks. This blue arrow is a suggestion, and you need to test if this LFC shape with your analysis. Like in (**h**) “Run Network with selected LFC” will show the next main network window.

c+h Test several LFC window

autoLFC – Network exploration

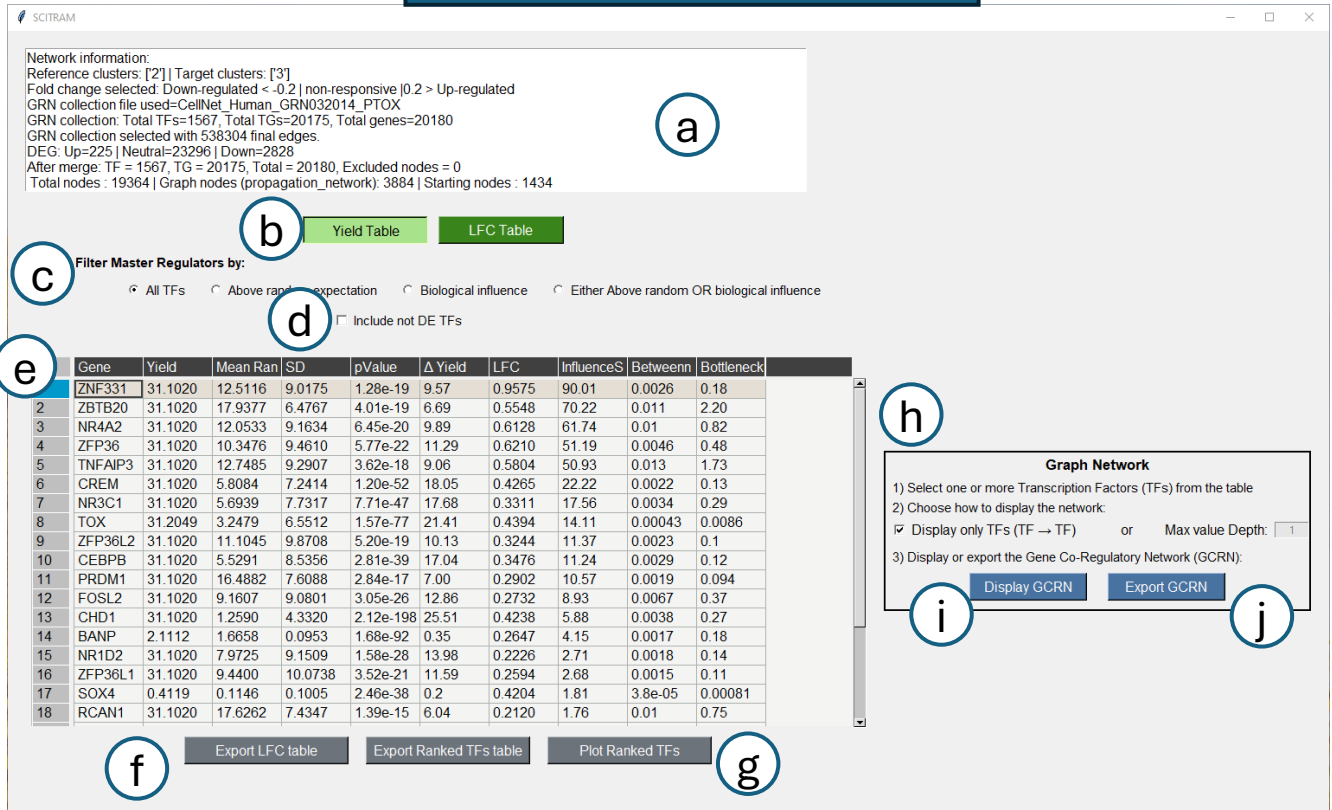
Networks by LFC						
	LFC	DEG	Nodes	Edges	Density	Component
1	0.1	5563	5968	45926	0.0013	1
2	0.2	3032	3877	25841	0.0017	1
3	0.3	1714	2743	15048	0.002	2
4	0.4	992	2074	9150	0.0021	4
5	0.5	568	1546	5440	0.0023	6
6	0.6	350	1213	3581	0.0024	5
7	0.7	210	900	2162	0.0027	4
8	0.8	117	640	1230	0.003	3
9	0.9	72	509	895	0.0035	3
10	1.00	45	384	559	0.0038	3

Blue row: Recommended LFC based on network connectivity stability

Run Network with Selected LFC

The main network window shows a list of ranked Transcription factors (TFs) based on the fraction of target genes under their control (in percent) over the whole reconstructed Gene regulatory network (GRN). This parameter is called “yield” or Master regulatory Index (MRI).

Main Network window



Network information:
Reference clusters: [2] | Target clusters: [3]
Fold change selected: Down-regulated < -0.2 | non-responsive [0.2 > Up-regulated
GRN collection file used=CellNet_Human_GRN032014_PTOX
GRN collection: Total TFs=1567, Total TGs=20175, Total genes=20180
GRN collection selected with 538304 final edges.
DEG: Up=225 | Neutral=23296 | Down=2828
After merge: TF = 1567, TG = 20175, Total = 20180, Excluded nodes = 0
Total nodes : 19364 | Graph nodes (propagation_network): 3884 | Starting nodes : 1434

Filter Master Regulators by:
☒ All TFs ☐ Above random expectation ☐ Biological influence ☐ Either Above random OR biological influence
☐ Include not DE TFs

	Gene	Yield	Mean Ran	SD	pValue	Δ Yield	LFC	InfluenceS	Betweenn	Bottleneck
1	ZNF331	31.1020	12.5116	9.0175	1.28e-19	9.57	0.9575	90.01	0.0026	0.18
2	ZBTB20	31.1020	17.9377	6.4767	4.01e-19	6.69	0.5548	70.22	0.011	2.20
3	NR4A2	31.1020	12.0533	9.1634	6.45e-20	9.89	0.6128	61.74	0.01	0.82
4	ZFP36	31.1020	10.3476	9.4610	5.77e-22	11.29	0.6210	51.19	0.0046	0.48
5	TNFAIP3	31.1020	12.7485	9.2907	3.62e-18	9.06	0.5804	50.93	0.013	1.73
6	CREM	31.1020	5.8084	7.2414	1.20e-52	18.05	0.4265	22.22	0.0022	0.13
7	NR3C1	31.1020	5.6939	7.7317	7.71e-47	17.68	0.3311	17.56	0.0034	0.29
8	TOX	31.2049	3.2479	6.5512	1.57e-77	21.41	0.4394	14.11	0.00043	0.0086
9	ZFP36L2	31.1020	11.1045	9.8708	5.20e-19	10.13	0.3244	11.37	0.0023	0.1
10	CEBPB	31.1020	5.5291	8.5356	2.81e-39	17.04	0.3476	11.24	0.0029	0.12
11	PRDM1	31.1020	16.4882	7.6088	2.84e-17	7.00	0.2902	10.57	0.0019	0.094
12	FOSL2	31.1020	9.1607	9.0801	3.05e-26	12.86	0.2732	8.93	0.0067	0.37
13	CHD1	31.1020	1.2590	4.3320	2.12e-198	25.51	0.4238	5.88	0.0038	0.27
14	BANP	2.1112	1.6658	0.0953	1.68e-92	0.35	0.2647	4.15	0.0017	0.18
15	NR1D2	31.1020	7.9725	9.1509	1.58e-28	13.98	0.2226	2.71	0.0018	0.14
16	ZFP36L1	31.1020	9.4400	10.0738	3.52e-21	11.59	0.2594	2.68	0.0015	0.11
17	SOX4	0.4119	0.1146	0.1005	2.46e-38	0.2	0.4204	1.81	3.8e-05	0.00081
18	RCAN1	31.1020	17.6262	7.4347	1.39e-15	6.04	0.2120	1.76	0.01	0.75

Graph Network
1) Select one or more Transcription Factors (TFs) from the table
2) Choose how to display the network:
☒ Display only TFs (TF → TF) or Max value Depth: 1
3) Display or export the Gene Co-Regulatory Network (GCRN):

Export LFC table **Export Ranked TFs table** **Plot Ranked TFs**

- a) Network Information:** This panel shows information about the parameters used for the analysis and it provides information concerning the reconstructed GRN (number of nodes, target genes, and number of TFs as starting nodes).
- b)** There are two options to show in the main table (**f**): the “Yield”, and the “LFC” table.
- c) Filter Transcription Factors.** This option allows to change the display of the table (**e**). Either display all TFs, or select those having values “Above random expectation” (only the TFs having yields higher than the randomized yield). Users can also select either to display the TFs with “biological influence” (explained below) or select “Either random OR biological influence” to maximize the TFs enriched by these two filters.
- d) Include no DE TFs:** It corresponds to display all TFs even if they are not differentially expressed.

- e) The Main Table, is interactive, allowing to change the ranking by performing a double-click on the header. Users can also copy/paste the content of the table. This table is composed by the following columns:

Gene: In this table this corresponds to the transcription factors (TF).

Yield: Percentage of genes under the control of the associated TF (direct or Indirect connections).

Mean Random Yield: average yield obtained over multiple randomized GRNs (20 randomizations by default).

SD: Standard deviation associated to the randomized yields.

P-value: Confidence assessed based on the multiple randomized GRNs (z-test).

Delta yield: the difference between the Yield and the mean random yield.

LFC: Differential expression (Log2) corresponding to the TF.

Influence score: A composite metric quantifying a node's regulatory impact by integrating its multi-depth propagation reach (yield_d1/d2/d3) with LFC-weighted influence.

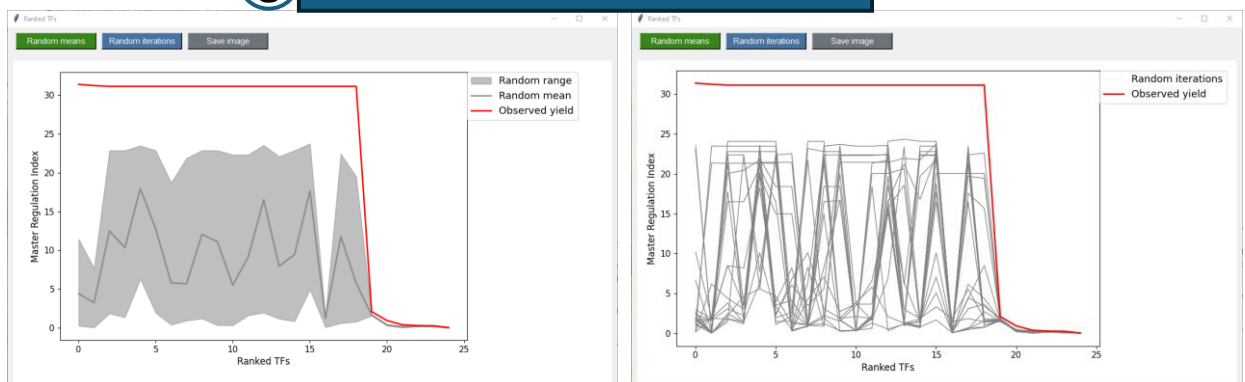
Betweenness: A centrality measure indicating how often a node lies on shortest paths between other nodes, reflecting its role as a mediator in information flow.

Bottleneck: A topology score combining betweenness and node degree to highlight highly connected nodes that also act as critical bridges in the network.

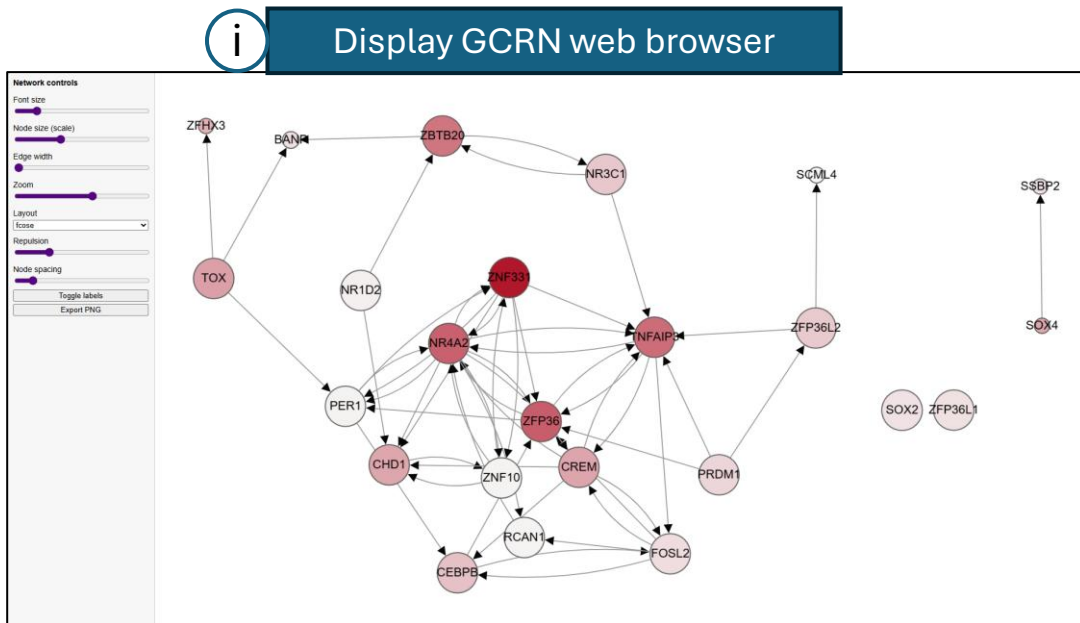
- f) Export the yield or LFC table, the export table will be the same as you see in the window, using the same filters and rankings.
- g) Plot Ranked TFs. The TFs with delta yield positive will be selected and ranked by yield, the mean and standard deviation range will be shown. You can see in this window 3 buttons, "random means" by default, "random iterations" to see each iteration and you can save this figure.

g

Plot ranked TFs supp window



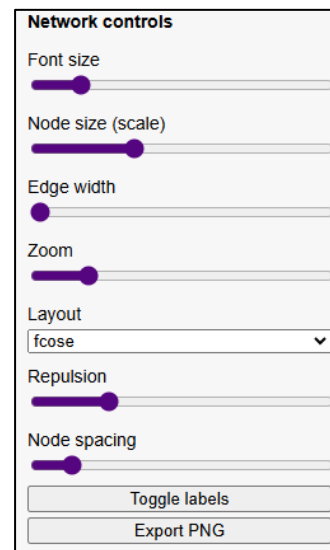
h) **Graph Network:** SCITRAM does not display the whole reconstructed GRN, but rather the regulomes linked to the master TFs predicted by the transcription regulation modeling. To graph or export your network, first you need to select the TFs in the table. By default, SCITRAM proposes to display the TF co-regulatory network (TF → TF) such that the regulatory relationship between the selected master TFs is displayed. Otherwise, users can obtain the complete graph with the interactions of the selected TF and the next neighbors to do this de select the box selector and use the “Max valued pth”, 1 is the first neighbors, 2 the seconds, etc (see one example below). To show in a web browser the figures you need to use the button (i) or the (j) to export in “.cx2” format, this format could be imported in cytoscape, a complete tool that requires additional installation (<https://cytoscape.org/>).



One web browser, one page will be open to show the network from the TFs selected (you can re open the html files going to the directory “Network”) At the left you have a panel to modify the network style:

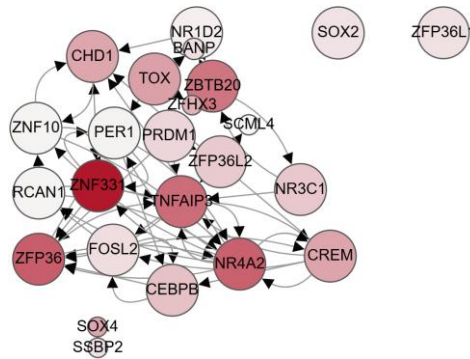
The font size could be higher or lower, also the size of the nodes; the edge line could be thicker, or you can do a zoom in the image. You can choose between different layouts, showed in the next section. For some layouts like cose and fclose, the repulsion could be increased to separate the nodes in a dynamic way. For the other layouts, the node spacing do the same work in a fixed way. Finally, you can hide the labels and export the image to png format.

The nodes could be manipulated in the space with the mouse and doing one click, more information will be displayed (yield and LFC values). Node colors corresponds to the LFC from white to red (min to max), and the size for the yield (min to max).

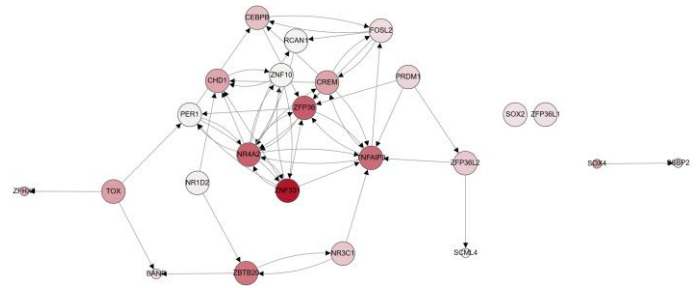


Layouts:

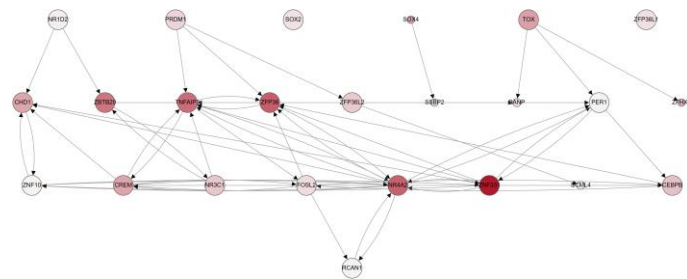
Cose:



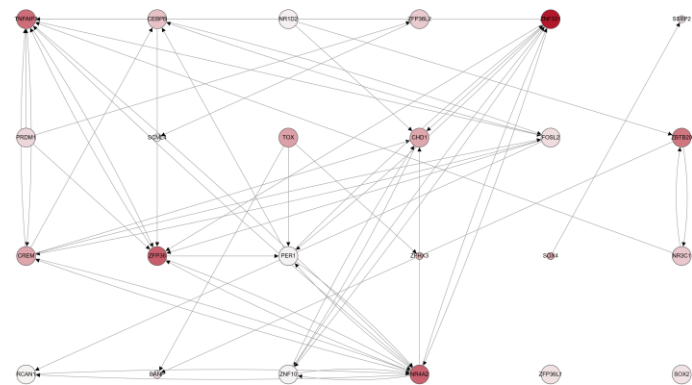
Fcose:



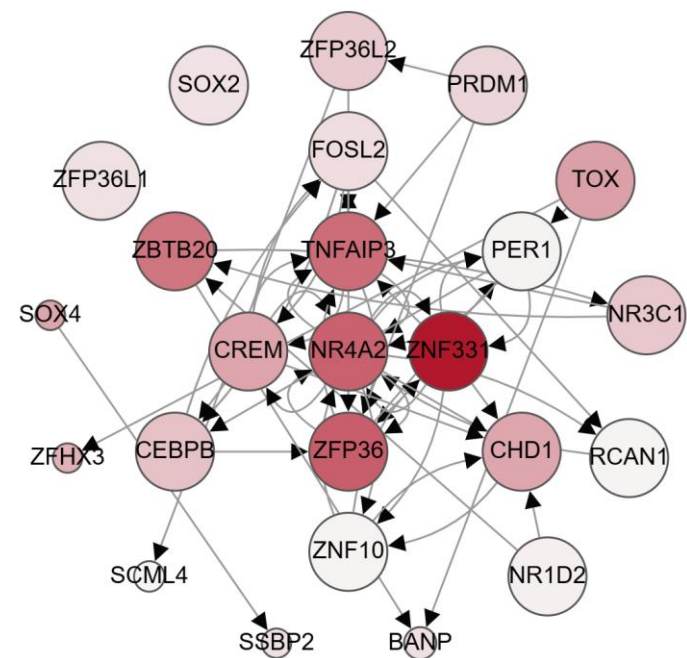
Breaththfirst:



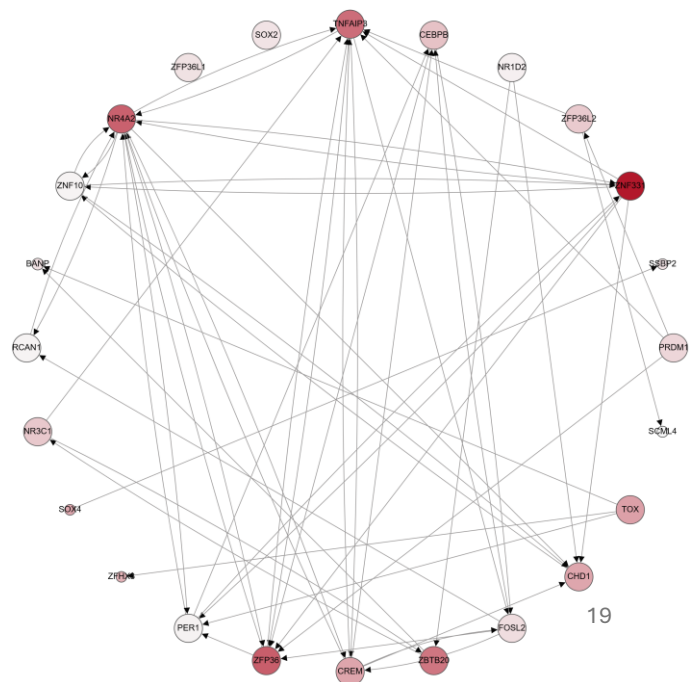
Grid:



concentric:



circle:



Network selecting TFs with delta yield positive and
Max depth =1:

